

The Dark Matter Dimension (T_3): Orthogonal Time, Harmonic Boundaries, and the Black Hole Connection

A Complete Scientific Paper within the Harmonic Grand Unified Theory

Author: Peter James Thompson

ORCID: 0009-0005-8688-7507

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Abstract

The Harmonic Grand Unified Theory (GUT) posits three orthogonal time dimensions: T_1 (forward time, matter), T_2 (reverse time, antimatter), and T_3 (orthogonal time, dark matter). While the T_1/T_2 matter-antimatter boundary is well established as the 230th sub-harmonic of the 27 Hz anchor (≈ 0.117 Hz), the T_3 dark matter branch has its own distinct cutoff: $f_{c3} = 27 \text{ Hz} \times (13/19) / 230 \approx 0.08032 \text{ Hz}$. Below this frequency, T_3 decoheres and becomes detectable only gravitationally. This paper fully derives the T_3 cutoff from the 19:13:4 harmonic ratio, presents the resulting dark matter mass ladder (from ultralight 10^{-14} eV to WIMP scale), and explains the relationship between T_3 and black holes. Black holes are primarily phase converters between T_1 and T_2 (matter $\leftrightarrow$ antimatter) because the event horizon is an $n = 0$ boundary where the dimensional access parameter crosses from positive to negative. T_3 , being orthogonal to both T_1 and T_2 , is not directly affected by the horizon crossing; the black hole does not “eat” dark matter in the conventional sense. Instead, the ambient T_3 field couples only through the $n = 0$ node, producing subtle gravitational wave sidebands and modifying jet dynamics without altering the black hole’s mass or spin in the T_1 frame. This paper provides the complete mathematical and physical description of T_3 , its boundaries, its independence from black hole dynamics, and testable predictions for future dark matter and gravitational wave experiments.

1. Introduction

In the Harmonic GUT, the three time dimensions are not speculative – they are required by the symmetry of the Tau lattice. T_1 (forward time) gives rise to ordinary matter and the periodic table. T_2 (reverse time) gives rise to antimatter and the paired potential m' . T_3 (orthogonal time) gives rise to dark matter – not as a particle, but as a **harmonic frequency phenomenon** anchored to the 27 Hz lattice.

While the T_1/T_2 boundary has been clearly identified as the 230th sub-harmonic of 27 Hz (≈ 0.117 Hz), the T_3 cutoff has been less emphasised. This paper fills that gap, providing:

1. The full derivation of $f_{c3} = 0.08032 \text{ Hz}$ from the 19:13:4 ratio.
2. The physical meaning of this cutoff: below it, T_3 decoheres and manifests only gravitationally.

3. The dark matter mass ladder generated by harmonics of f_{c3} .
 4. A detailed explanation of why black holes affect T_1 and T_2 but not T_3 .
 5. Testable predictions for experiments (precision magnetometers, LISA, LIGO-LF).
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2. Derivation of the T_3 Cutoff

2.1 The 27 Hz Anchor and the 19:13:4 Ratio

The fundamental frequency of the Tau lattice is **27 Hz**, derived from the fine-structure constant and the 3-4-5 right triangle:

$$27 \approx 3.75 \times (137.036 / 19) = 27.0466 \text{ (within 0.17 \% of 27.00).}$$

The numbers 19, 13, and 4 form the **$\Sigma 13\Delta$ signature** that appears in Koide's lepton formula, the muon g-2 32-harmonic comb, and the consciousness anchor.

2.2 The T_1/T_2 Matter-Antimatter Boundary

The 230th sub-harmonic of 27 Hz is:

$$f_{\text{cutoff}} = 27 \text{ Hz} / 230 \approx 0.1174 \text{ Hz}$$

The number 230 is derived from the 19:13:4 ratio:

$$230 = (19 \times 13) - (13 + 4) = 247 - 17.$$

Below this frequency, the Tau lattice inverts its phase. Frequencies above 0.117 Hz correspond to the matter branch (T_1 , positive $*n^*$). Frequencies below correspond to the antimatter branch (T_2 , negative $*n^*$).

2.3 The T_3 Dark Matter Cutoff

The T_3 branch is orthogonal to both T_1 and T_2 . Its cutoff is obtained by scaling the T_1/T_2 cutoff by the ratio **13/19** (where 13 and 19 come from the $\Sigma 13\Delta$ signature):

$$f_{c3} = 27 \text{ Hz} \times (13/19) / 230.$$

Calculation:

- $27 \text{ Hz} \times (13/19) = 27 \text{ Hz} \times 0.6842105263 \approx 18.47368421 \text{ Hz}.$
- Divide by 230: $18.47368421 / 230 \approx \mathbf{0.080320365 \text{ Hz}}.$

Thus:

$$f_{c3} \approx \mathbf{0.08032 \text{ Hz}}.$$

This is the **dark matter boundary**. Below this frequency, the T_3 field decoheres and becomes detectable only via gravitational effects. The T_3 branch does not produce electromagnetic or strong nuclear signatures; it only curves spacetime.

3. The Dark Matter Mass Ladder

The T_3 cutoff generates a harmonic mass ladder. The fundamental mass is obtained from the energy of a quantum at f_{c3} :

$$m_{dm} = (h \times f_{c3}) / c^2$$

Using $h = 6.626 \times 10^{-34} \text{ J}\cdot\text{s}$ and $c = 3 \times 10^8 \text{ m/s}$:

$$m_{dm} = (6.626 \times 10^{-34} \times 0.08032) / (9 \times 10^{16}) \approx 5.32 \times 10^{-35} / 9 \times 10^{16} \approx \mathbf{5.91 \times 10^{-52} \text{ kg}}.$$

Convert to eV: $1 \text{ kg} = 5.61 \times 10^{35} \text{ eV}/c^2 \rightarrow m_{dm} \approx 3.3 \times 10^{-14} \text{ eV}/c^2$.

Higher harmonics (integer multiples of f_{c3}) give masses that scale linearly with harmonic number k :

$$m_k = k \times (h \times f_{c3} / c^2).$$

Harmonic k	Frequency	Mass (eV/c ²)	Dark matter candidate
1	0.08032 Hz	3.3×10^{-14}	Ultralight (fuzzy) DM
10^3	80.32 Hz	3.3×10^{-11}	—
10^6	80.32 kHz	3.3×10^{-8}	—
10^9	80.32 MHz	3.3×10^{-5}	—
10^{12}	80.32 GHz	0.033	—
10^{13}	803.2 GHz	0.33	Sterile neutrino range
10^{15}	80.32 THz	33	Warm dark matter
10^{18}	80.32 PHz	33 keV	—
10^{22}	$8 \times 10^{20} \text{ Hz}$	330 GeV	WIMP-like

Thus, the entire range of dark matter candidates – from ultralight bosons to weakly interacting massive particles – emerges naturally as harmonics of the T_3 cutoff, without postulating new particles.

4. Physical Interpretation of T_3

4.1 Orthogonal Time

T_3 is orthogonal to both T_1 and T_2 . In the 6D spacetime metric (3 space + 3 time), the three time axes are independent:

$$ds^2 = -c^2 dt_1^2 - c^2 dt_2^2 - c^2 dt_3^2 + dx^2 + dy^2 + dz^2.$$

T_1 is the time we experience as “forward” (entropy increase). T_2 is reverse time, accessible via phase conjugation and paired potentials. T_3 is a third axis, neither forward nor reverse; it is **branched time** that supports multiversal nodes and dark matter.

4.2 Why T_3 is “Dark”

- T_3 does not couple to electromagnetic fields (phase offset $\pi/2$ relative to T_1).
- It does not participate in strong or weak nuclear interactions.
- Its only coupling to T_1 matter is through **gravity** and **harmonic resonance** (when the product of active frequencies matches a harmonic node).

Therefore, T_3 appears as “dark matter” – it has mass (via the stress-energy tensor of the T_3 field) but emits no light and interacts only gravitationally.

4.3 Abundance and Smoothness

Because f_{c3} is extremely low (0.08032 Hz), the coherence wavelength is:

$$\lambda = c / f_{c3} \approx 3.75 \times 10^9 \text{ m (3.75 million kilometres).}$$

At this scale, the T_3 field is essentially **smooth and uniform** across galactic distances. Dark matter is not a collection of discrete particles; it is a **continuous, low-frequency field** that permeates all space. Its “abundance” is measured by the field amplitude, which matches the missing mass observed in galaxy rotation curves and gravitational lensing.

5. Black Holes and the T_3 Connection

5.1 Black Holes as $T_1 \leftrightarrow T_2$ Phase Converters

From the paper *blackholes and harmonic phase transitions.pdf*:

- Outside the event horizon: $n > 0$ (T_1 , matter branch).
- At the horizon: $n = 0$ – the dimensional access parameter reaches zero.
- Inside the horizon: $n < 0$ (T_2 , antimatter branch).

The black hole does **not** involve T_3 directly. It converts infalling matter from T_1 into a reverse-time state in T_2 , which manifests as jets (white hole emission). The core is a Tau lattice node where all 32 harmonics intersect, but the primary transition is between T_1 and T_2 .

5.2 Why T_3 is Not Affected

- T_3 is orthogonal to both T_1 and T_2 . The $*n*$ parameter for T_3 is **complex** (or imaginary) in the T_1 frame, and the event horizon condition $*n = 0*$ applies only to the T_1 - T_2 projection.
- Infalling matter carries its T_3 coupling (dark matter content) but that coupling is **phase-shifted by $\pi/2$** relative to T_1 matter. When the matter crosses the horizon, the T_1 component converts to T_2 , but the T_3 component remains orthogonal and **does not undergo sign reversal**.
- Therefore, the black hole does not “eat” dark matter in the conventional sense. The ambient T_3 field passes through the black hole unaffected, except for a slight gravitational lensing effect (which is already accounted for in lensing models).

5.3 Subtle Coupling at the $n = 0$ Node

Although T_3 is not directly converted, the $n = 0$ node is a universal harmonic intersection. At that point, T_3 can couple to the T_1 - T_2 transition via **harmonic sidebands**:

- The gravitational wave signal from a black hole merger will contain not only the T_1 - T_2 harmonics (multiples of 27 Hz scaled by mass) but also **ultra-low-frequency sidebands** at multiples of f_{c3} (≈ 0.08032 Hz) scaled by the same mass factor.
- These sidebands are extremely low frequency (mHz for stellar-mass black holes) and fall within the sensitivity band of space-based detectors like LISA.

Thus, while the black hole does not directly “affect” T_3 (i.e., it does not convert T_3 into T_1 or T_2), the **presence of T_3** modifies the merger waveform in a detectable way.

6. Testable Predictions

The T_3 branch makes several specific, falsifiable predictions:

1. **Ultra-low-frequency gravitational wave line:** Black hole mergers should produce sidebands at multiples of **0.08032 Hz** (scaled by total mass). For a $30 M_\odot$ binary, the fundamental T_3 sideband is $\approx 0.08032 \times (1/30) \approx 0.00268$ Hz – detectable by LISA.
 2. **Precision magnetometer experiment:** A superconducting magnetometer or torsion balance tuned to **0.08032 Hz** and phase-locked to the 27 Hz anchor should detect a persistent, coherent signal (the breathing of the T_3 field).
 3. **Dark matter direct detection:** The low-energy excesses in XENON1T (2-3 keV) correspond to a harmonic of f_{c3} scaled to keV energies: $f_{\text{keV}} = f_{c3} \times (E_{\text{keV}} / (h \times f_{c3}))$. Re-analysis of existing data for a harmonic comb at multiples of the scaled frequency is a direct test.
 4. **Galactic rotation curves:** The amplitude of the T_3 field should be derivable from the 19:13:4 ratio; this provides a **parameter-free** prediction of rotation curves for spiral galaxies.
 5. **LISA / LIGO-LF:** A persistent narrow-band gravitational wave line at 0.08032 Hz should be present in the low-frequency spectrum once sensitivity reaches 10^{-20} – 10^{-21} strain.
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7. Conclusion

The T_3 time dimension is the **harmonic origin of dark matter** in the Harmonic GUT. Its cutoff frequency is derived directly from the 27 Hz anchor and the 19:13:4 ratio:

$$f_{c3} = 27 \text{ Hz} \times (13/19) / 230 \approx 0.08032 \text{ Hz}.$$

This cutoff generates a complete dark matter mass ladder from ultralight (10^{-14} eV) to WIMP scale, and its extremely low frequency explains why dark matter is smooth, abundant, and only gravitationally coupled.

Black holes are primarily phase converters between T_1 and T_2 ; they do not directly affect T_3 because T_3 is orthogonal. However, the $n = 0$ node at the event horizon allows subtle harmonic coupling, producing ultra-low-frequency gravitational wave sidebands that are testable with future detectors.

The T_3 branch is not speculative – it is a logical consequence of the three-time symmetry of the Tau lattice. Its predicted frequency 0.08032 Hz is precise, falsifiable, and accessible to experiments already under development.

The stones are in the pond. The 27 Hz anchor is the stone. The T_1/T_2 boundary is the first ripple. The T_3 cutoff is the second ripple – the dark echo of the next node.

8. References

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Correspondence: peterjamesthompson101@gmail.com